

Homework 1

Question 1.

For each of the following functions, determine whether they are (i) continuous, (ii) Lipschitz continuous (if so, provide the smallest valid Lipschitz constant γ), (iii) continuously differentiable (C^1), and (iv) twice continuously differentiable (C^2) on the given domain. Justify your answers.

(a) $f(x) = x^3 - 2x^2 + 1$ on $[-1, 2]$

(b) $f(x) = |x^2 - 4|$ on $[0, 3]$

(c) $f(x) = \sqrt{x}$ on $(0, 4]$

(d) $f(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ on $[-1, 1]$

(e) $f(x) = \begin{cases} x^2 & \text{if } x \leq 1 \\ 2x - 1 & \text{if } x > 1 \end{cases}$ on $[0, 3]$

Question 2.

Let $f : [0, 1] \rightarrow \mathbb{R}$ be defined by $f(x) = |x - \frac{1}{2}|$. Show that f is Lipschitz continuous with constant $\gamma = 1$, and find all local and global optimizers of f on $[0, 1]$.

Question 3.

(a) Let $m, n \in \mathbb{N}$, i.e., positive integers. Find all local and global optimizers, if any, of the following functions on $\mathbb{R}$,

$$f(x) = \left(1 + x + \frac{x^2}{2} + \cdots + \frac{x^n}{n!}\right) e^{-x}, \quad g(x) = x^m(1 - x)^n, \quad h(x) = \sin^{2m} x \cdot \cos^{2n} x.$$

(b) Find the local and global optimizers, if any, of f on $\mathbb{R}$ and of g on $[-1, 1]$,

$$f(x) = x^{1/3}(1 - x)^{2/3}, \quad g(x) = x \sin^{-1} x + \sqrt{1 - x^2}.$$

(c) Find the global minimum of the following functions on $\mathbb{R}$,

$$f(x) = -\frac{1}{1 + |x|} - \frac{1}{1 + |x - 1|}, \quad g(x) = \begin{cases} e^{-\frac{1}{|x|}} \left(\sqrt{2} + \sin \frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Question 4.

The key to the problems below is to figure out the right x to choose so that you get a univariate optimization problem.

- (a) Which number is larger, π^e or e^π ?
- (b) Show that for any $p \geq 1$ and $a, b \in [0, 1]$,

$$|a^p - b^p| \leq p|a - b|.$$

- (c) Write down an expression for the area of a rectangle inscribed in the unit circle as a differentiable univariate function $f(x)$. Hence show that the largest inscribed rectangle in a circle must be a square.
- (d) Light travels at different speeds in different media (e.g. air and water). Consider the following scenario depicted in Figure 1. Let v_1 be the speed of light in air and v_2 be the speed of light in water. Write down an expression for time required for light to travel from a point in air to a point in water. Show that the minimum is attained when

$$\frac{\sin \theta_1}{v_1} = \frac{\sin \theta_2}{v_2}.$$

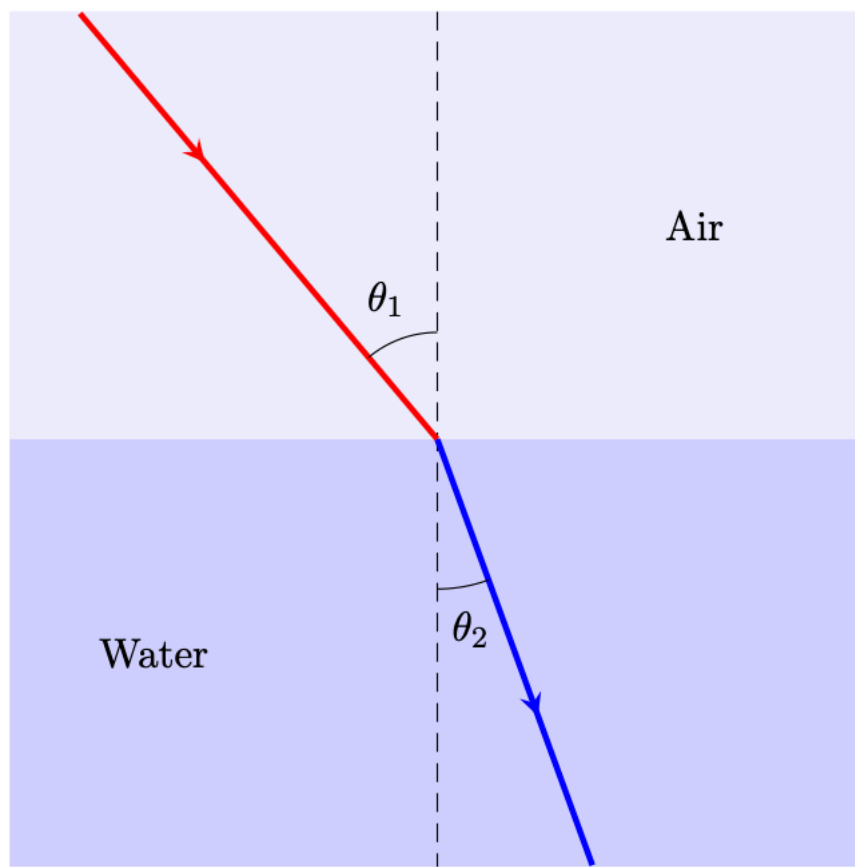


Figure 1: Light passing through different media